

Analysis of some numerical linear algebra algorithm

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1 Neumann series approximation

The principle of Neumann series is from the identity

$$(I + P + P^2 + \dots + P^k)(I - P) = I - P^{k+1} \quad (1)$$

Under the assumption, $\lim_{k \rightarrow \infty} P^k = 0$, the approximation is true

$$(I - P)^{-1} = \sum_{k=0}^{\infty} P^k. \quad (2)$$

The right hand side of (2) is known as Neumann series. From the book [1], we have some related theorems.

Theorem 1. *Let $P \in \mathbb{C}^n$ and $\lim_{k \rightarrow \infty} P^k = 0$. Then $I - P$ is invertible and (2) holds. Furthermore, the error can be evaluated as*

$$\|I + P + P^2 + \dots + P^k - (I - P)^{-1}\| \leq \frac{\|P\|^{k+1}}{1 - \|P\|} \quad (3)$$

Proof. see theorem 4.2 in [1]. □

2 QR decomposition

The QR representation can be expressed due to this theorem.

Theorem 2. *Let $X \in \mathbb{C}^{n \times p}$ have rank p . Then X can be written in unique form*

$$X = QR, \tag{4}$$

where Q is an $n \times p$ orthogonal matrix and R is upper triangular matrix with positive diagonal elements.

Proof. see Theorem 4.24 in [1]. □

References

- [1] G.W. Stewart. *Matrix Algorithms: Volume 1: Basic Decompositions*. Society for Industrial and Applied Mathematics, 1998.